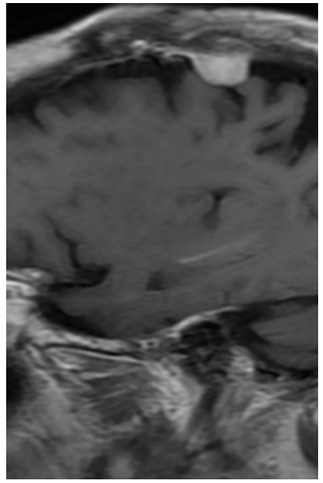
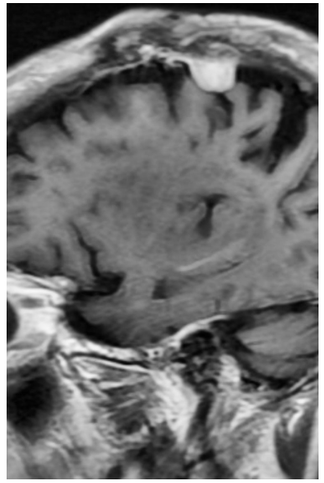
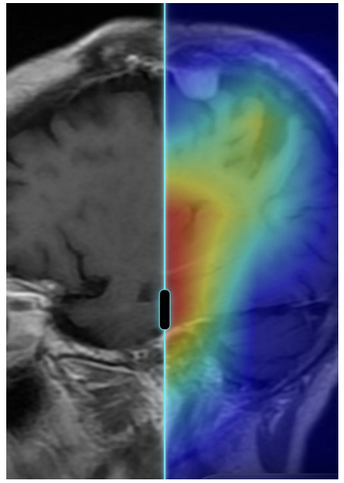




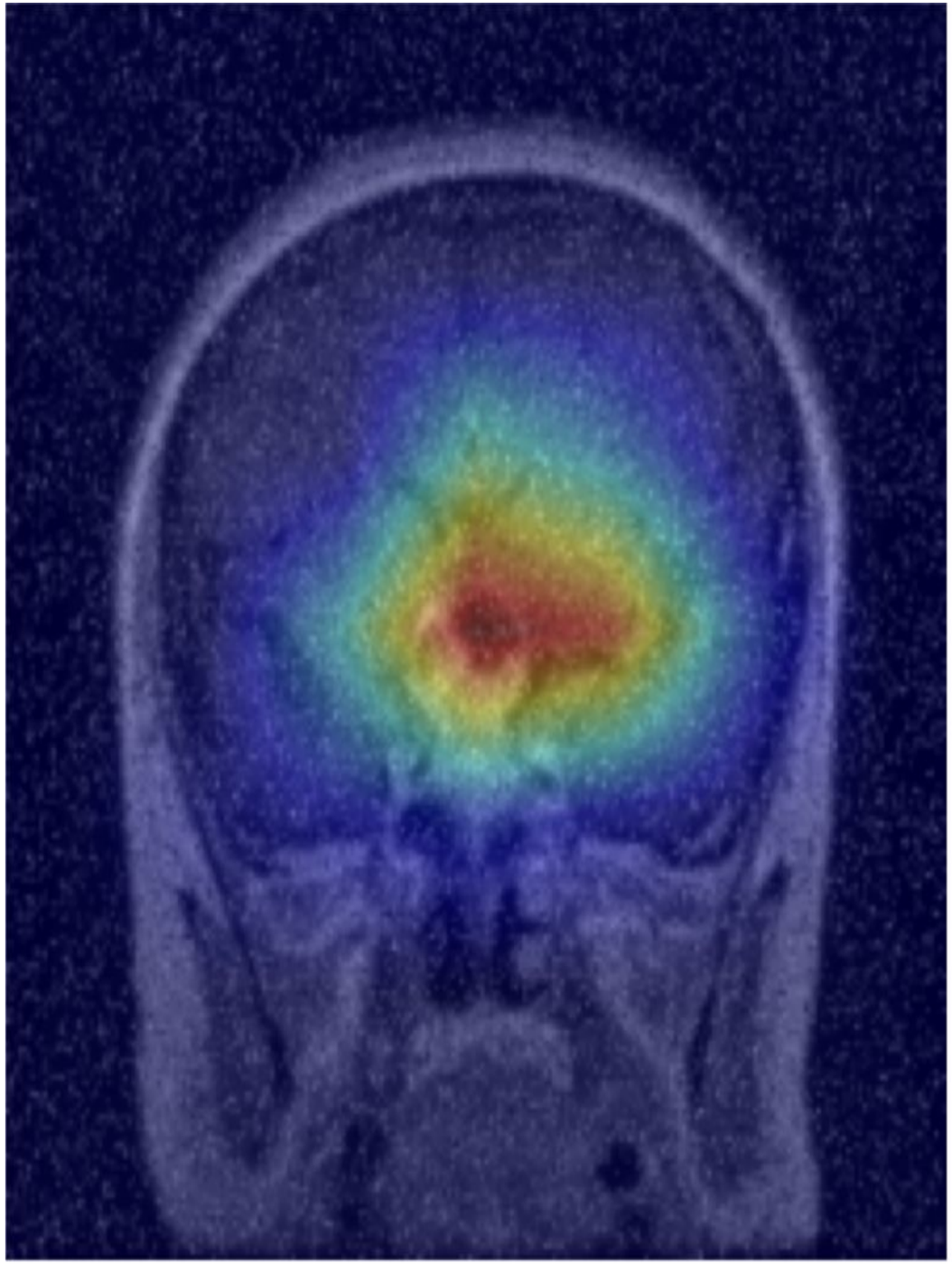


Neural Nexus

Next-Gen AI Clinical Control Center

Transforming brain tumor diagnostics through explainable AI, unified workflows, and clinical interpretability

Submitted By:



The Clinical Challenge

Black Box Problem

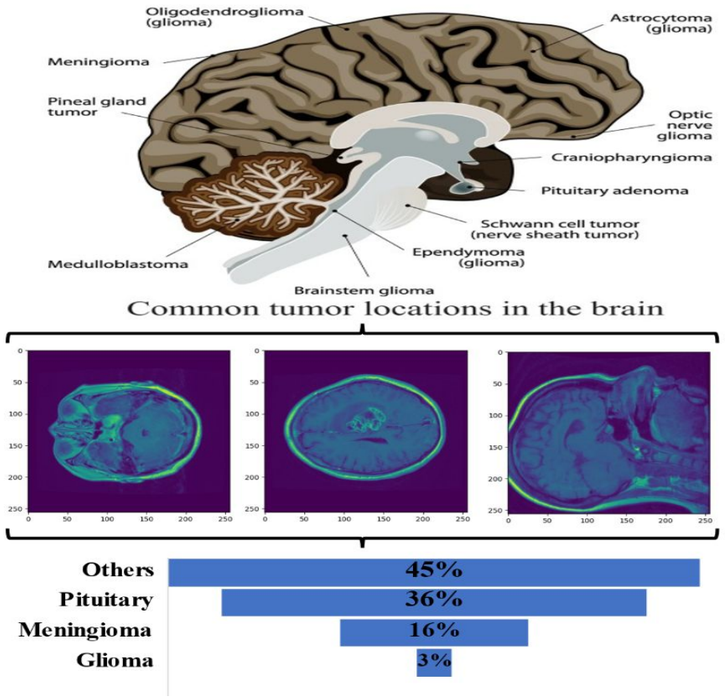
AI outputs without explanations create trust barriers in clinical settings

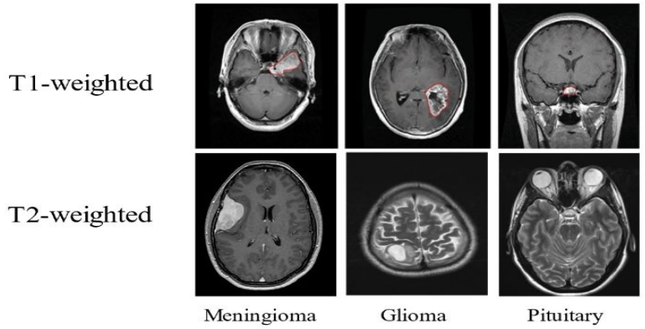
Class Imbalance

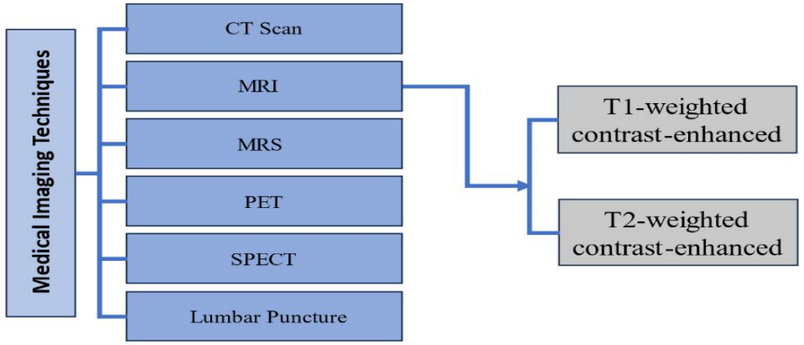
It was hard to determine rare case diseases of tumor. Ex: Pituitary

Complexity

MRI images are high standard, noisy, and have deep pathological features.







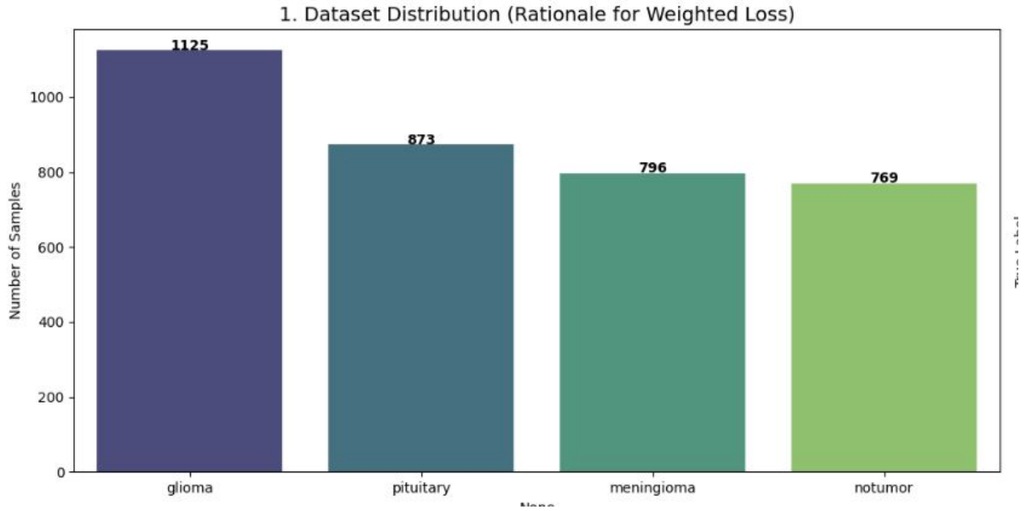
The Unified Solution

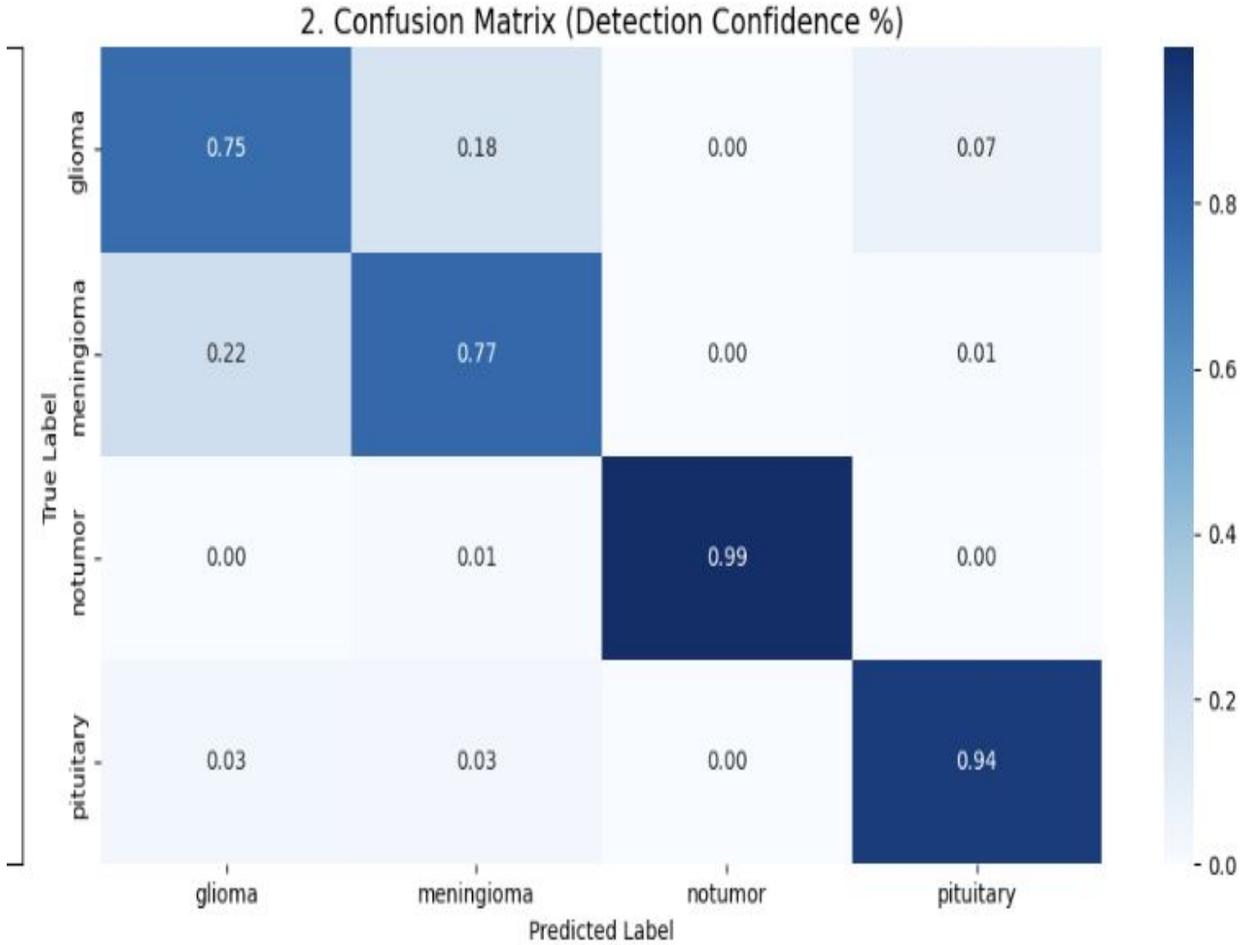
Single Integrated Platform

MRI scans, AI insights, and clinical explanations unified in one interface

- Instant risk calculation
- Real-time interpretability
- No tool switching required

Reduces diagnosis time while improving decision clarity





ML Architecture: The Diagnostic Core

01

ResNet-50 with Transfer Learning

Pre-trained model adapted for limited medical data

02

Four-Class Classification

Glioma, meningioma, pituitary tumor, no tumor

03

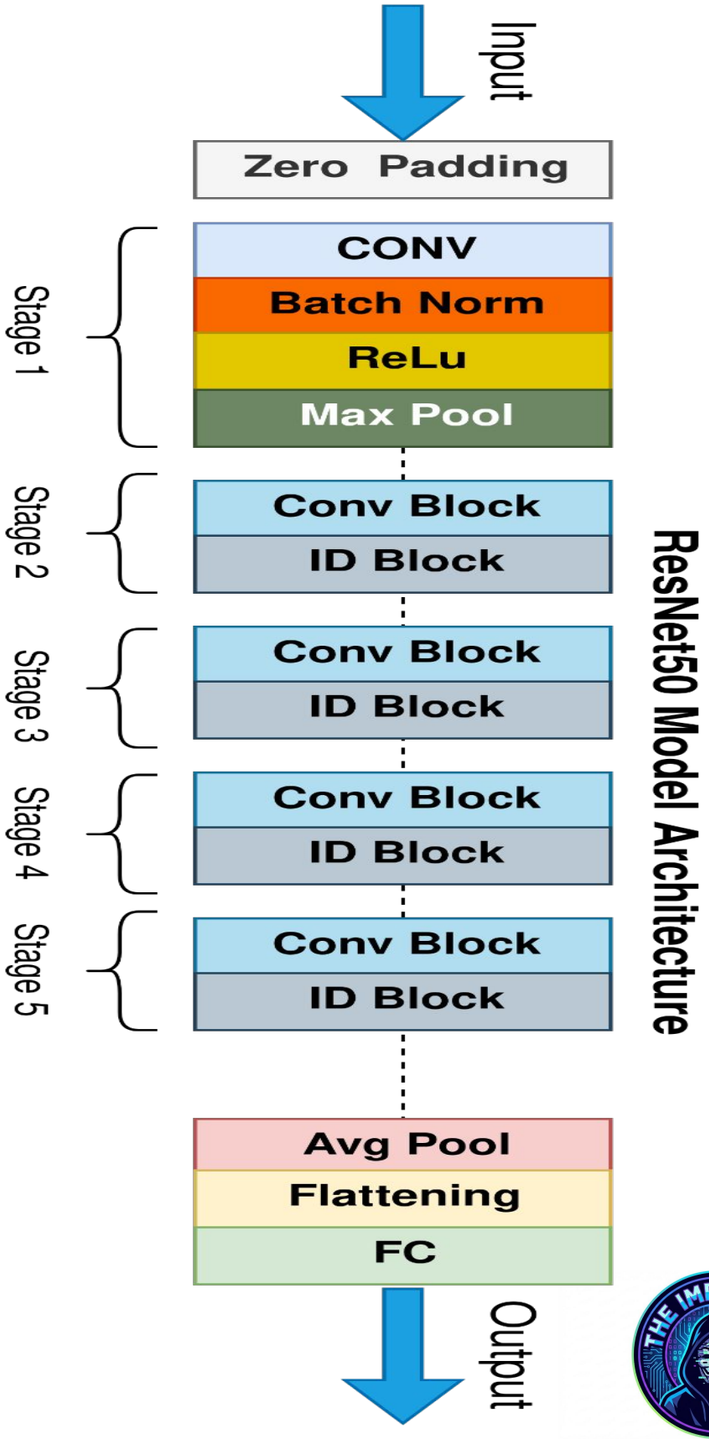
Image Preprocessing

Normalization and contrast enhancement improve accuracy, Resizing (224×224)

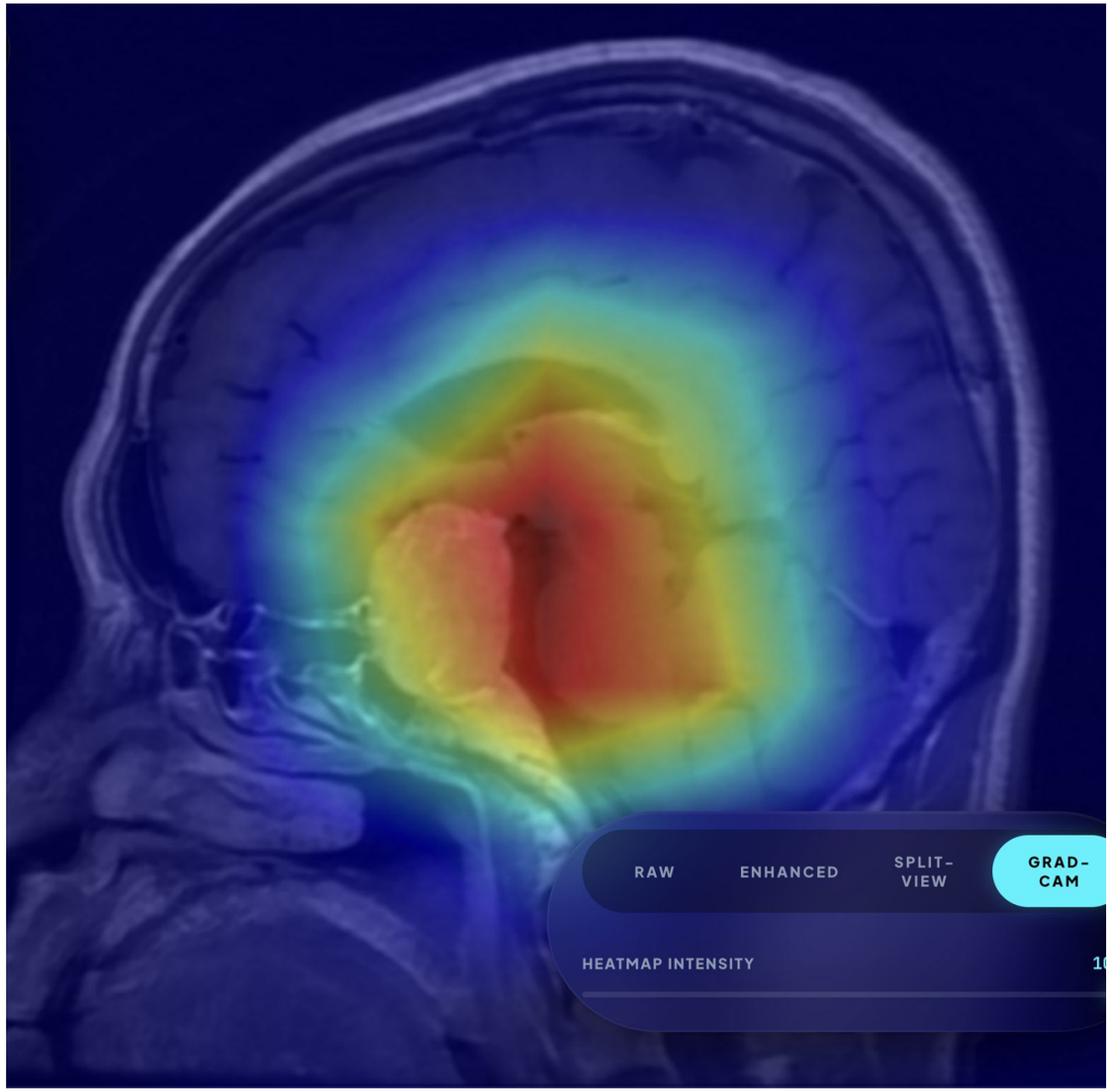
04

Clinical Pattern Recognition

Augmentation Pipeline and GradCam



XAI and Interpretability

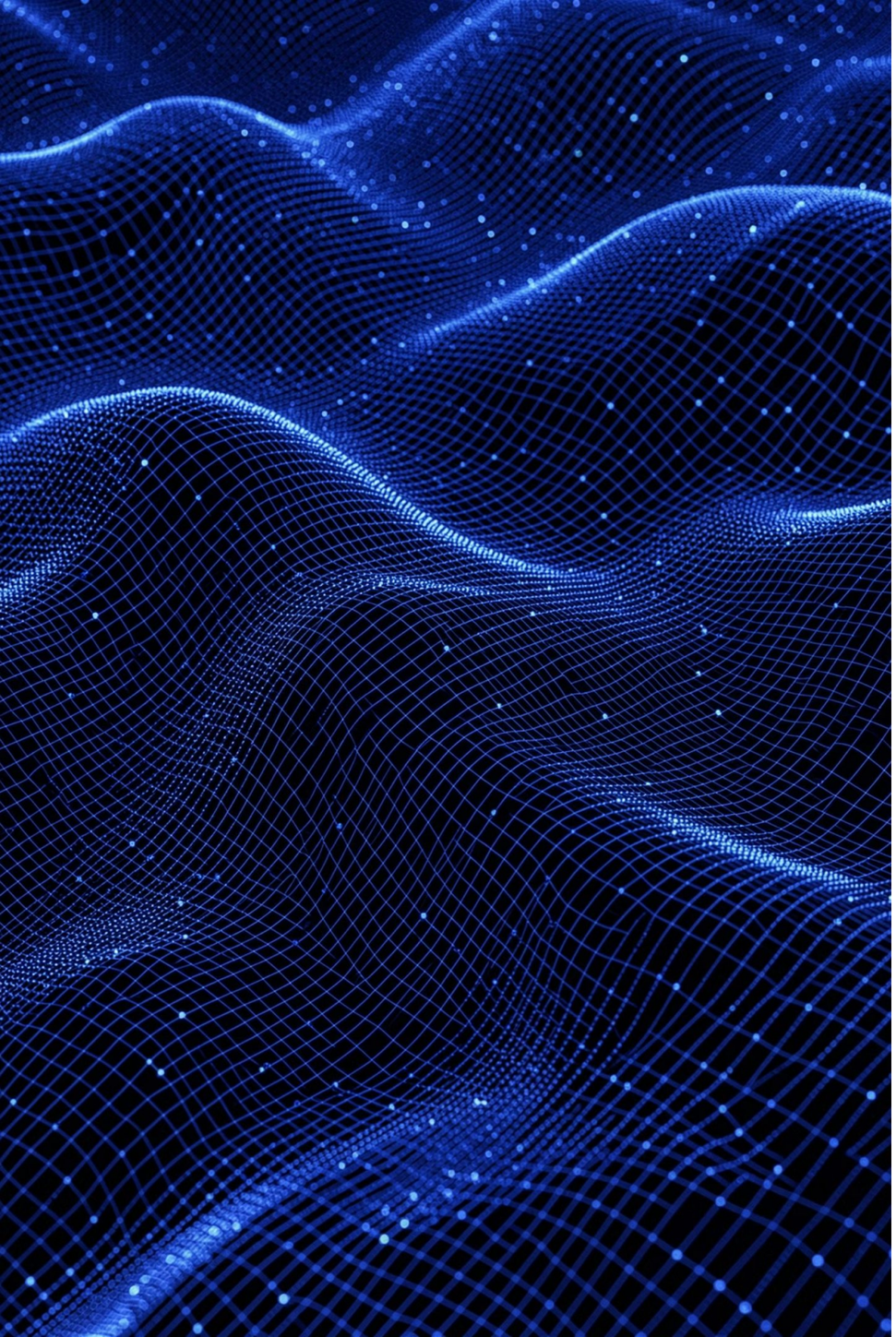


Custom Grad-CAM Methodology

Highlights MRI regions influencing predictions without external libraries

- Adjustable heatmap intensity
- Direct anatomical correlation
- Full system control

Transparency builds clinical trust



Integrating BioMistral

1

BioMistral-7B

Primary LLM for clinical text generation

2

Mistral-7B Fallback

Backup model ensures reliability

3

Clinical Narrative

Structured explanations replace raw probabilities

Transforms predictions into actionable insights: diagnosis explanation, risk evaluation, and next-step recommendations

Quantitative Clinical Risk Metrics

Tumor Irregularity

Entropy calculations indicate malignancy risk

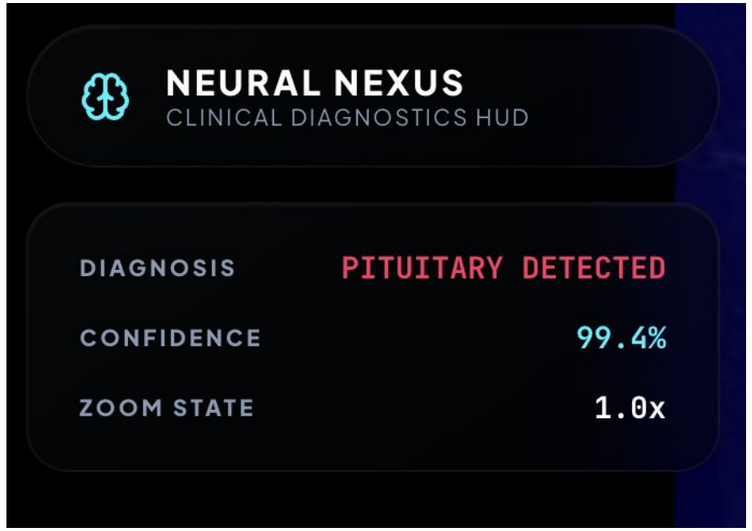
Activation Area

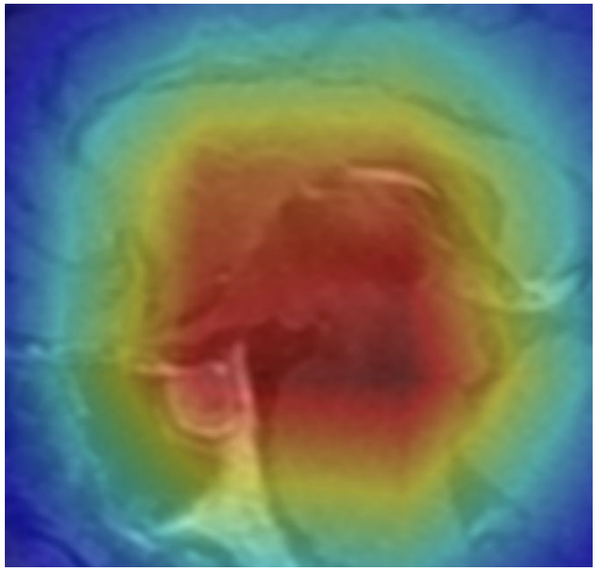
Brain volume affected measurement

Composite Score

0-100 severity index for clear communication

Moves beyond simple classification to meaningful clinical quantification



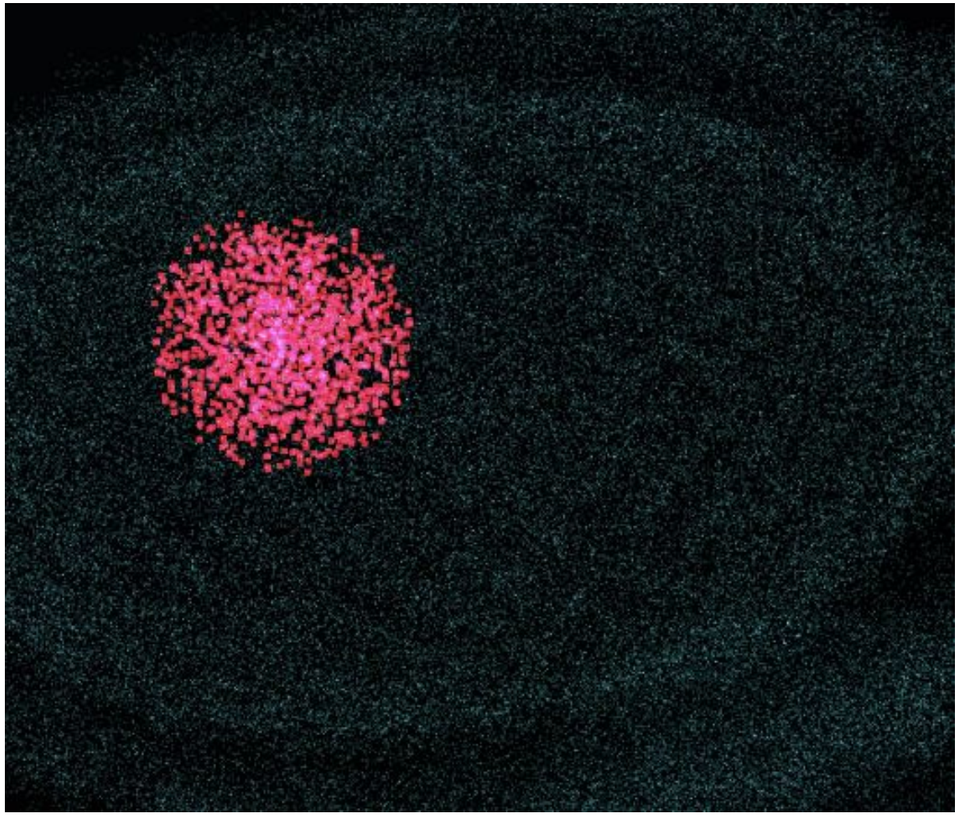


3D Spatial Copilot

Dynamic Integration

2D MRI slices synchronized with 3D brain model in real-time

- Click 3D model → navigate to exact slice
- Detected regions auto-highlight
- Particle-based visualization



User Experience Design

Glass-Style Interface

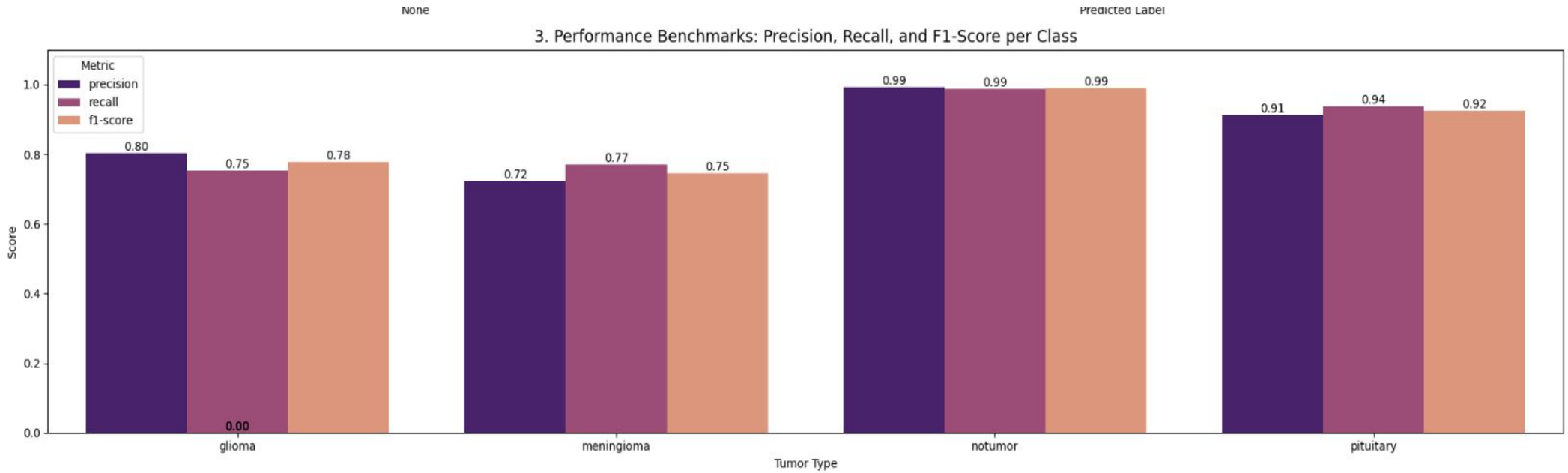
Dark backgrounds with clear highlights reduce eye strain during extended use

Unified Display

All critical information visible without page switching or tab navigation

Natural Interactions

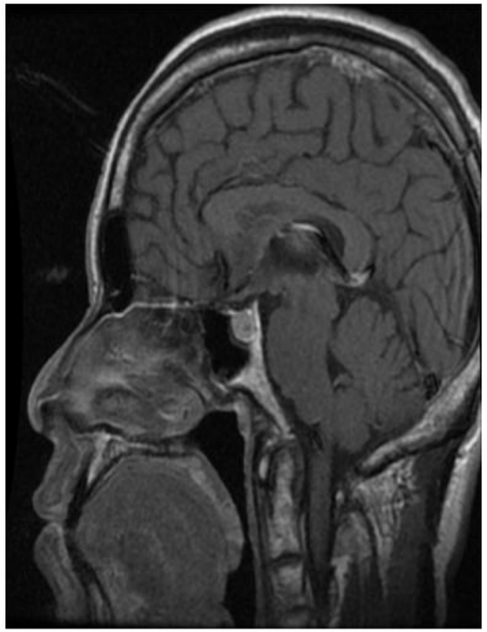
Smooth animations and responsive design create intuitive clinical workflow



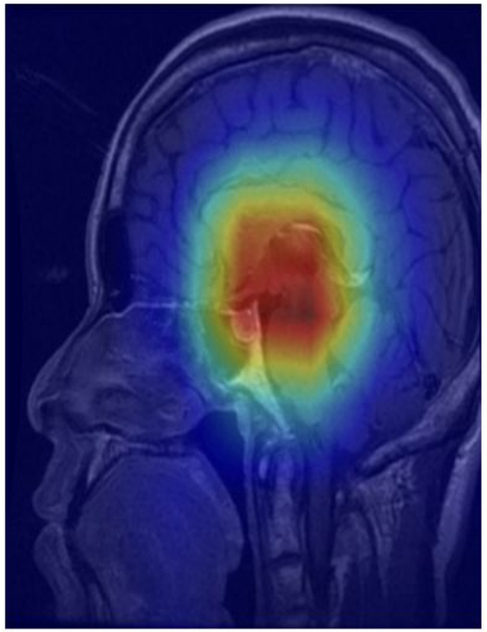
PRIMARY DIAGNOSIS: Pituitary

System Confidence: 99.44%

Original MRI Frame



AI Attention Map (Grad-CAM)



Probability Distribution Breakdown

Glioma:	0.14%
Meningioma:	0.11%
No Tumor:	0.31%
Pituitary:	99.44%

Conclusion and Roadmap

Current Deployment

Production-ready with Vercel frontend and Hugging Face backend services

Automated Reports

Complete clinical documentation generation

3D Overview of Brain

A 3d architecture to view along

Bayesian Confidence

Improved uncertainty quantification